

Notes: Duflo, Greenstone, Pande, and Ryan (Econometrica, 2018)
The Value of Regulatory Discretion: Estimates from Environmental
Inspections in India

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Institutional context	4
3.1	Environmental regulation in Gujarat	4
3.2	Penalties	4
3.3	Why discretion is ambiguous	4
4	Experimental design	5
4.1	Sample and randomization	5
4.2	Treatment	5
4.3	Why the treatment is deliberately bundled	5
5	Data and measurement	5
5.1	Administrative regulatory records	5
5.2	End-line survey	5
5.3	Key outcomes	6
6	Empirical strategy	6
6.1	Intent-to-treat regressions	6
6.2	Pollution and compliance regressions	6
7	Structural model	6
7.1	Why a model is needed	6
7.2	Targeting stage	6
7.3	Regulator objective	7
7.4	Penalty stage	7
8	Tables and figures	8
8.1	Table I: Structure of Penalty Stage Actions and Table II: Regulatory Interactions During Experiment	8
8.2	Table III: End-Line Pollution and Compliance on Treatments and Figure 1: The Effect of Treatment on Pollution Distribution	9
8.3	Figure 2: Model Fit to Inspections and Pollution and Figure 3: Regulatory Targeting of Extreme Polluters	10
8.4	Figure 4: Actions of the Regulatory Machine and Plant at Each Node and Table IV: Multinomial Logit Model of Action Choice Conditional on State	11
8.5	Table V: Estimates of Plant Utility Parameters and Figure 5: Value of Environmental Regulation for Plants	12

8.6	Table VI: Estimates of Targeting Stage Parameters and Figure 6: Sensitivity of Targeting Parameters to Moments	13
8.7	Figure 7: Observed and Optimal Targeting Rules and Figure 8: Value of Discretion: Abatement by Information Regime and Budget Constraint	14
9	Main findings by theme	14
9.1	Finding 1: the experiment sharply increased inspections	14
9.2	Finding 2: more inspections produced more detected violations and citations	14
9.3	Finding 3: more inspections did not generate many more costly penalties	15
9.4	Finding 4: average pollution effects were small	15
9.5	Finding 5: random inspections missed extreme violators	15
9.6	Finding 6: the regulator targets aggressively but with weak information	15
9.7	Finding 7: discretion has high value at low budgets	15
10	Identification and interpretation	16
10.1	What the randomized experiment identifies	16
10.2	What the experiment alone does not identify	16
10.3	Why the structural model is credible in this setting	16
10.4	Why treatment effects are small despite more inspections	16
11	Economic intuition	16
11.1	Why targeted inspections can dominate random inspections	16
11.2	Why information quality matters	16
11.3	Why more inspections may have diminishing returns	17
12	Relation to broader literature	17
13	Strengths and limitations	17
13.1	Strengths	17
13.2	Limitations	17
14	Short summary	18
15	Conclusion	18
15.1	Core conclusion	18
15.2	Incremental contribution revisited	18
15.3	Policy implication	18
15.4	Takeaway	19

1 Big picture

- **Question.** Does giving regulators discretion over which plants to inspect improve or weaken environmental enforcement when inspection resources are scarce?
- **Setting.** The paper studies environmental regulation in Gujarat, India, where the Gujarat Pollution Control Board (GPCB) regulates highly polluting industrial plants under strict air and water pollution standards.
- **Core problem.** Developing-country environmental law may be strict on paper, but enforcement is constrained by limited inspection budgets and imperfect information about which plants are dirty.
- **Experiment.** The authors worked with GPCB to increase inspections for a random subset of highly polluting plants. The treatment both increased inspection resources and removed discretion over the added inspections by assigning them randomly.
- **Headline experimental result.** The treatment more than doubled inspection rates and increased detected violations and citations, but it did not meaningfully reduce average pollution emissions. It only modestly increased compliance.
- **Puzzle.** Why did many more inspections have weak pollution effects? The answer is not that inspections were ignored: treatment and control inspection reports entered the same regulatory process and received similar follow-up conditional on findings.
- **Proposed explanation.** The random extra inspections mostly found moderate violators. The regulator's discretionary inspections, though infrequent, were already disproportionately finding the most extreme violators.
- **Structural result.** A structural model shows that discretion in targeting is valuable: at the status quo inspection budget, regulator-chosen inspections induce about three times as much abatement as the same number of random inspections.
- **Policy takeaway.** The problem is not simply too much discretion or too few inspections. Discretion has value when regulators possess useful private information, but the regulator's information is noisy. Better monitoring technology can make discretion much more productive.

2 Incremental contribution of the paper

- **Experimental contribution.** The paper provides rare field-experimental evidence on how environmental inspections affect plant behavior and pollution.
- **Rules-versus-discretion contribution.** It does not ask only whether inspections matter; it asks whether regulator discretion over targeting matters.
- **Unbundling contribution.** Because the treatment bundles more inspection resources with less discretion, the paper develops a structural model to separate the value of resources from the value of targeting discretion.
- **Regulatory-process data contribution.** The authors code nearly 10,000 regulator-plant documents over five years, including inspections, pollution readings, warnings, penalties, and plant responses.

- **Measurement contribution.** The paper combines administrative enforcement data with an independent end-line survey of plant pollution and abatement costs.
- **Revealed-preference contribution.** The penalty-stage model uses plants' compliance choices to infer the private cost of regulation, including formal penalties and informal costs such as disruption or bribes.
- **Policy-counterfactual contribution.** The model simulates inspection regimes that the experiment did not directly test: uniform inspections, discretionary inspections with current information, discretionary inspections with perfect information, and a hybrid regime like the experiment.
- **Broader state-capacity contribution.** The paper shows that autonomy can be valuable for bureaucrats when they have local information, but that the value of autonomy depends on the quality of their information.

3 Institutional context

3.1 Environmental regulation in Gujarat

GPCB monitors roughly 20,000 industrial plants. Plants in the highest pollution category are subject to routine inspection mandates. Small plants are supposed to be inspected once per year, while medium and large plants are supposed to be inspected more frequently.

Interpretation: The law is not weak. The key constraint is implementation: which plants are inspected, how often, and whether information from inspections produces credible penalties.

3.2 Penalties

Plants found in violation can face several escalating actions:

- citation or warning letters;
- mandates to install or upgrade abatement equipment;
- bank guarantees or performance bonds;
- temporary plant closure;
- utility disconnection.

Interpretation: Penalties can be severe when applied, but the probability of facing a costly penalty is low. This combination is central for the paper's interpretation: inspections affect behavior mainly if they raise expected penalty costs enough to justify costly abatement.

3.3 Why discretion is ambiguous

Discretion can hurt enforcement if inspectors avoid dirty plants, shirk, or are captured by regulated firms. But discretion can help enforcement if regulators have local signals about pollution that are not visible to the econometrician. The paper's key insight is that these two views must be separated empirically.

4 Experimental design

4.1 Sample and randomization

The experiment covered 960 red-category small- and medium-scale industrial plants in Ahmedabad, Surat, and Valsad. Treatment assignment was randomized within region and audit-experiment strata. The inspection treatment was cross-randomized with the audit reform experiment studied in Duflo et al. (2013).

4.2 Treatment

Treatment plants were assigned at least one annual initial inspection and up to four inspections per year. After an initial inspection, plants were randomly assigned to quarterly follow-up inspections with probability 0.66. The treatment changed two things at once:

- **More resources.** Treatment plants received many more inspections.
- **Less discretion over marginal inspections.** The extra inspections were randomly assigned rather than targeted by regulators.

The treatment did not change pollution standards or penalty rules. It also did not change the regulator’s discretion over its existing inspection budget.

4.3 Why the treatment is deliberately bundled

A clean two-by-two experiment would separately vary inspection budgets and discretion. The authors could not implement all arms. Instead, the experiment tested the de jure policy of regular inspections for all plants, and the structural model later separates the role of more inspections from the role of targeting.

5 Data and measurement

5.1 Administrative regulatory records

The authors code 9,624 GPCB documents on interactions between the regulator and sample plants. These documents are classified into regulator actions and plant actions.

- **Regulator actions:** inspect, warn, punish, accept.
- **Plant actions:** comply, ignore.

The coded documents are linked into chains of interaction. A chain begins with an inspection and continues as the regulator and plant alternate moves until the regulator accepts the plant’s compliance or takes no further action.

5.2 End-line survey

An independent end-line survey measured pollution and abatement costs after the experiment. Surveyors collected pollution readings, plant abatement expenditures, and other operating information. Results were not used by GPCB for enforcement.

5.3 Key outcomes

- **Inspection outcomes:** assigned inspections, actual inspections, initial inspections.
- **Regulatory actions:** readings collected, readings above standard, citations, warnings, closure directions, bank guarantees, equipment mandates, utility disconnections.
- **Plant costs:** capital abatement spending and maintenance spending.
- **Pollution:** standardized pollutant readings relative to each pollutant’s standard.
- **Compliance:** indicator that a pollutant reading is below its regulatory standard.

6 Empirical strategy

6.1 Intent-to-treat regressions

The experimental analysis estimates effects of treatment assignment. A generic plant-level regression is

$$Y_i = \alpha + \beta \text{InspectionTreatment}_i + \Gamma_s + \varepsilon_i,$$

where Γ_s are randomization-strata controls such as region and audit-sample indicators. The coefficient β is the causal effect of being assigned to the inspection treatment.

6.2 Pollution and compliance regressions

For pollution and compliance, the unit is plant-by-pollutant. The paper estimates

$$Y_{ip} = \alpha + \beta_1 T_i^I + \beta_2 T_i^A + \beta_3 (T_i^I \times T_i^A) + \Gamma_s + \varepsilon_{ip},$$

where T_i^I is inspection treatment, T_i^A is audit treatment, and standard errors are clustered at the plant level.

Interpretation: The interaction term matters because inspections and third-party audits both generate pollution information. If the two information channels substitute for each other, the interaction can be positive even when each main effect reduces pollution.

7 Structural model

7.1 Why a model is needed

The experiment answers the effect of *random additional inspections*. It does not directly answer what would have happened if the same number of additional inspections had been targeted by the regulator. The structural model is designed to evaluate these counterfactual policies.

7.2 Targeting stage

Plant j has latent pollution in period m :

$$\log P_{jm} = \phi_0 + \phi_1' X_j + u_{1j} + u_{2jm}.$$

Here X_j are observable plant characteristics, u_{1j} is a persistent pollution signal observed by the regulator but not by the econometrician, and u_{2jm} is a shock known to the plant but not the regulator.

Abatement reduces pollution proportionally:

$$\log \tilde{P}_{jm} = \log P_{jm} + \phi_2 \text{Run}_{jm}, \quad \phi_2 < 0.$$

The plant runs abatement equipment if the expected reduction in penalty costs exceeds maintenance cost:

$$\text{Run}_{jm}^* = 1 \left\{ I_j \left[V_0(P_{jm}) - V_0(\tilde{P}_{jm}) \right] > c_j \right\}.$$

Interpretation: Inspections matter through expected penalties. A dirty plant is more likely to abate because an inspection gives it more to lose if it does not run abatement equipment.

7.3 Regulator objective

The regulator chooses an inspection targeting rule to maximize expected pollution abatement subject to an inspection budget. The targeting rule maps observed plant characteristics, treatment status, and the regulator's pollution signal into an expected annual number of inspections. In the estimated model, the rule takes a probit-link form with a maximum inspection level and a shift parameter.

Economic message: The shape of the targeting rule is the key object for policy. A flat rule means equal inspections for all plants. A steep rule means the regulator concentrates inspections on plants with high pollution signals.

7.4 Penalty stage

After an inspection, the regulator and plant alternate actions. The regulator can inspect again, warn, punish, or accept. The plant can comply or ignore. The plant's dynamic problem trades off immediate abatement or compliance costs against the probability and cost of future regulatory penalties.

Interpretation: The penalty-stage value function $V_0(P)$ summarizes the expected discounted cost to a plant of entering the regulatory process after a pollution reading P .

8 Tables and figures

8.1 Table I: Structure of Penalty Stage Actions and Table II: Regulatory Interactions During Experiment

THE VALUE OF REGULATORY DISCRETION 2131

TABLE I
STRUCTURE OF PENALTY STAGE ACTIONS^a

Round	Regulatory Action				Plant Action		N (7)	% Left (8)
	Inspect (1)	Warn (2)	Punish (3)	Accept (4)	Ignore (5)	Comply (6)		
1	100.0	0.0	0.0	0.0			7423	100.0
2					99.6	0.4	7423	
3	1.0	9.5	2.2	87.3			7423	100.0
4					92.8	7.2	941	
5	23.3	4.8	5.3	66.6			941	12.7
6					91.1	8.9	314	
7	18.8	11.8	9.9	59.6			314	4.2
8					83.5	16.5	127	
9	21.3	5.5	18.1	55.1			127	1.7
10					82.5	17.5	57	
11	26.3	3.5	10.5	59.6			57	0.8
12					87.0	13.0	23	
13	26.1	4.3	8.7	60.9			23	0.3
14					77.8	22.2	9	
15+	16.7	8.3	0.0	75.0	100.0	0.0	9	0.1
Total without inspections	0.0	4.6	1.6	42.7	50.2	0.9	7824	
Total	31.0	3.2	1.1	29.4	34.6	0.6	25,217	

^aThe table reports actions taken by the regulatory machine and by the firm in the penalty stage using administrative data. Figure 5 defines actions and their payoffs and Table II maps them to regulatory documents. Each of columns 1–6 gives the probability, within that row, of the party moving at that round when taking the action indicated in the column header. Column 7 gives the total number of actions observed in that round and column 8 gives the percentage of penalty stages that continue up to at least that round. The penalty stage always starts with an inspection. Action rounds within the stage then alternate between actions of the regulatory machine and sample plants. The penalty stage ends when the machine accepts. Rounds after the 15th round are not shown and the row 15+ summarizes these rounds: 6 chains go at least 17 rounds and 4 chains go 19 rounds.

(a) Table I: Structure of Penalty Stage Actions

THE VALUE OF REGULATORY DISCRETION 2133

TABLE II
REGULATORY INTERACTIONS DURING EXPERIMENT^a

	Control	Treatment	Difference
<i>Panel A. Inspections by Treatment Status</i>			
Number of inspections assigned in treatment, annual	0	2.12	2.12***
	[0]	[0.57]	(0.026)
Total inspections, annual over treatment	1.40	3.11	1.71***
	[1.59]	[1.77]	(0.11)
Initial inspections, annual over treatment	1.28	2.79	1.50***
	[1.38]	[1.52]	(0.094)
Observations	480	480	
<i>Panel B. Perceived Inspections by Treatment Status</i>			
Perceived inspections, 2008	2.53	2.66	0.13
	[1.42]	[1.40]	(0.10)
Perceived inspections, 2009	2.78	3.16	0.38***
	[1.44]	[1.37]	(0.190)
Perceived inspections, 2010	2.92	3.62	0.71***
	[1.58]	[1.46]	(0.11)
Total perceived notices and closures received, 2010	0.27	0.30	0.025
	[0.64]	[0.70]	(0.048)
Observations	388	403	
<i>Panel C. Regulatory Actions by Treatment Status</i>			
Pollution reading ever collected at plant (=1)	0.38	0.60	0.21***
	[0.49]	[0.49]	(0.032)
Any pollution reading above limit at plant (=1)	0.34	0.55	0.22***
	[0.47]	[0.50]	(0.031)
Number of pollution readings above limit at plant	1.17	2.84	1.67***
	[2.58]	[3.67]	(0.20)
Total citations	0.15	0.35	0.20***
	[0.42]	[0.69]	(0.037)
Total water citations	0.046	0.12	0.071***
	[0.22]	[0.37]	(0.020)
Total air citations	0.021	0.042	0.021*
	[0.14]	[0.20]	(0.011)
Total closure warnings	0.094	0.17	0.077***
	[0.34]	[0.48]	(0.027)
Total closure directions	0.16	0.20	0.042
	[0.48]	[0.54]	(0.033)
Total bank guarantees	0.060	0.065	0.0042
	[0.27]	[0.25]	(0.017)
Total equipment mandates	0.027	0.040	0.013
	[0.19]	[0.23]	(0.014)
Total utility disconnections	0.040	0.042	0.0021
	[0.22]	[0.20]	(0.013)
Observations	480	480	

^aThe table shows differences in actual inspection rates (panel A), perceived inspection rates (panel B), and other regulatory actions (panel C) between the treatment and control groups of plants during the treatment period of approximately 2 years. The second and third columns show means with standard deviations given in brackets. The third column shows the coefficient from regressions of each variable on treatment, where each regression includes region fixed effects and a control for the audit sample. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

(b) Table II: Regulatory Interactions During Experiment

Read: Table I summarizes the regulatory-machine and plant-action chains. Most cases end quickly: after an initial inspection, the regulator often accepts or stops. A small number of chains continue for many rounds, and the probability of punishment rises in later rounds. Table II shows that the experiment was implemented strongly: treatment plants received about 1.71 more total inspections per year and 1.50 more initial inspections per year.

Economic message: The first table shows why the expected penalty from a single inspection can be limited: most chains terminate early. The second table shows the core experimental puzzle. The treatment substantially increased readings above limits and citations, but it did not significantly increase costly penalties such as closure directions, equipment mandates, or utility disconnections.

8.2 Table III: End-Line Pollution and Compliance on Treatments and Figure 1: The Effect of Treatment on Pollution Distribution

TABLE III
END-LINE POLLUTION AND COMPLIANCE ON TREATMENTS^a

	(1)	(2)	(3)	(4)
<i>Panel A. Plant-Level Costs</i>				
	<i>Capital Costs</i>		<i>Maintenance Costs</i>	
	(USD × 10 ³)	Any (=1)	(USD × 10 ³)	Any (=1)
Inspection treatment (=1)	-0.221 (0.453)	0.0213 (0.0344)	0.838* (0.499)	0.00974 (0.0224)
Plant characteristics	Yes	Yes	Yes	Yes
Audit experiment	Yes	Yes	Yes	Yes
Control mean	2.050	0.567	0.264	0.108
Observations	791	791	791	791
<i>Panel B. Plant-by-Pollutant Level Pollution</i>				
	<i>Pollution</i>		<i>Compliance</i>	
Inspection treatment (=1)	-0.105 (0.0839)		0.0366* (0.0213)	
Audit treatment (=1)	-0.187** (0.0849)		0.0288 (0.0258)	
Audit × inspection treatment (=1)	0.286** (0.142)		-0.0365 (0.0353)	
Control mean	0.682		0.614	
Observations	4168		4168	

^aThe table shows intent-to-treat effects of inspection treatment assignment on plant costs and pollution outcomes. Panel A shows regressions for plant costs estimated at the plant level. Costs are divided into capital and maintenance costs based on descriptions of each expenditure (see Appendix A). Cost amounts are in thousands of U.S. dollars (USD). Capital costs, which are reported as lump sum in the survey, are amortized to an equivalent constant annual expenditure (using an interest rate of 20% and a 10-year equipment

(a) Table III: End-Line Pollution and Compliance on Treatments

2136

DUFLO, GREENSTONE, PANDE, AND RYAN

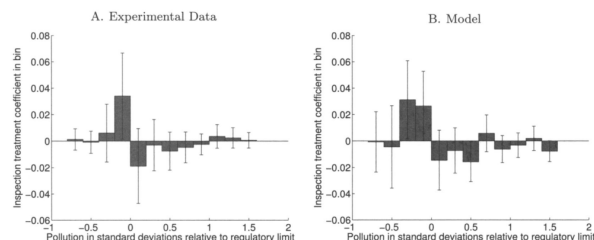


FIGURE 1.—The effect of treatment on pollution distribution. The figures report coefficients on the inspection treatment assignment from regressions of dummies for a pollution reading being in a given bin, relative to the regulatory standard, on inspection treatment, audit treatment, inspection × audit treatment, a dummy for being audit-eligible, and region fixed effects. Part (A) reports coefficients from such regressions on the experimental data and (B) reports coefficients from the same regressions run on model-generated data using the constrained model estimates of Table VI. Pollution readings are standardized by subtracting the regulatory standard for each pollutant and dividing by the pollutant's standard deviation; bins are 0.2 standard deviations wide and centered at the regulatory standard shown by the vertical line. Each plant has multiple pollutant observations and regressions are run pooled for all pollutants together. The “whiskers” show 95% confidence intervals for the inspection treatment coefficient.

(b) Figure 1: The Effect of Treatment on Pollution Distribution

Read: Table III reports intent-to-treat effects on abatement costs, pollution, and compliance. Inspection treatment has no meaningful effect on capital abatement expenditures, raises maintenance spending modestly, reduces pollution by about 0.105 standard deviations but not significantly, and increases compliance by about 3.7 percentage points. Figure 1 shows where the compliance effect comes from: treatment shifts mass from just above the standard to just below the standard.

Economic message: The treatment moved some marginal violators into compliance, but it did not induce large reductions in average pollution. This pattern is consistent with plants responding enough to meet the standard but not enough to substantially reduce emissions below the standard.

8.3 Figure 2: Model Fit to Inspections and Pollution and Figure 3: Regulatory Targeting of Extreme Polluters

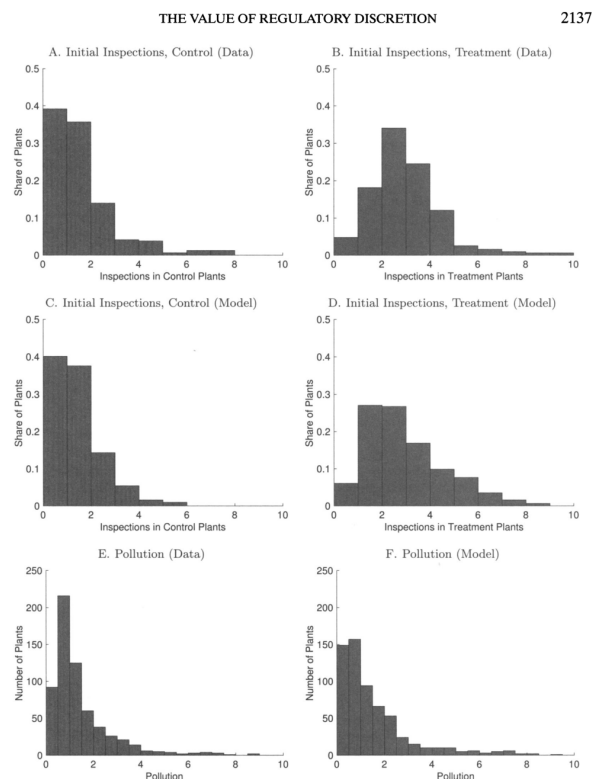


FIGURE 2.—Model fit to inspections and pollution. The figure compares the distributions of inspections and pollution in the model to those in the experimental data. Panels (A)–(D) show the distributions of the annual inspection rate (i.e., inspections per year). Inspections include only initial inspections and not followups. Panels (A) and (B) give the distributions in the data in the control and treatment groups, respectively, using administrative records of inspection reports. Panels (C) and (D) give the same distributions in the model.

(a) Figure 2: Model Fit to Inspections and Pollution

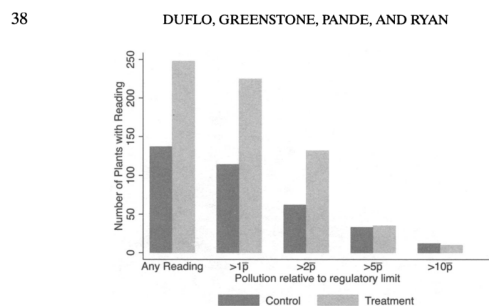


FIGURE 3.—Regulatory targeting of extreme polluters. The figure shows the number of plants with pollution readings either taken or that fall in various bins, relative to the regulatory standard, during the first year of intervention for the control and treatment groups, respectively. The first pair of bars shows the number of plants that had at least one pollution reading taken. The remaining four pairs show the number of plants with at least one reading above the standard ($>1\bar{p}$), more than 2 times the standard ($>2\bar{p}$), more than 5 times the standard ($>5\bar{p}$), and more than 10 times the standard ($>10\bar{p}$).

(b) Figure 3: Regulatory Targeting of Extreme Polluters

Read: Figure 2 compares the distributions of inspections and pollution in the data and model. The model matches the broad shape of inspection and pollution distributions. Figure 3 shows that treatment increased the number of plants with any reading and with readings above the standard, but did not increase the number of extreme violators above five or ten times the standard.

Economic message: Random inspections find many moderate violations but not many severe violators. This supports the paper's interpretation: the status quo discretionary inspections were already targeted toward the dirtiest plants.

8.4 Figure 4: Actions of the Regulatory Machine and Plant at Each Node and Table IV: Multinomial Logit Model of Action Choice Conditional on State

2142

DUFLO, GREENSTONE, PANDE, AND RYAN

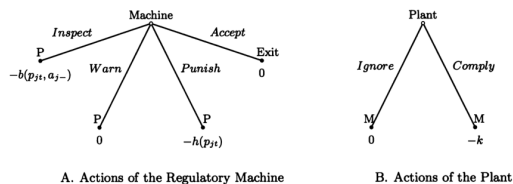


FIGURE 4.—Actions of the regulatory machine and plant at each node. The terminal nodes give the payoffs in each round for the plant. The penalty stage begins with an inspection where the regulatory machine (M) observes p_{jt} . The machine can take four actions. If M inspects, M gets a new signal of pollution and the plant may have to offer a bribe with payoff $-b(p_{jt}, a_{j-})$. If M warns, there is no cost to the plant. If M punishes, the plant faces a cost $-h(p_{jt})$. After each of these moves, the plant ignores or complies and M moves again. If M accepts, the stage ends.

(a) Figure 4: Actions of the Regulatory Machine and Plant at Each Node

THE VALUE OF REGULATORY DISCRETION

2147

TABLE IV
MULTINOMIAL LOGIT MODEL OF ACTION CHOICE CONDITIONAL ON STATE*

Party to move:	Regulatory Machine			Plant
	Inspect (1)	Warn (2)	Punish (3)	Comply (4)
Lagged regulatory actions				
Warn, lag 1	0.33 (0.23)	-2.05*** (0.32)	-2.10*** (0.31)	-0.23 (0.30)
Punish, lag 1	1.80*** (0.23)	-2.22*** (0.56)	-0.53* (0.30)	1.29*** (0.26)
Lagged plant actions				
Firm: Comply, lag 1	-1.80*** (0.32)	-1.03** (0.47)	-0.82** (0.37)	-0.53 (0.66)
Last observed pollution reading				
0-1x	-0.38 (0.23)	-0.25 (0.16)	0.052 (0.24)	-0.18 (0.38)
1-2x	-0.20 (0.16)	0.55*** (0.098)	0.37** (0.18)	0.39* (0.23)
2-5x	-0.17 (0.17)	0.84*** (0.10)	0.70*** (0.17)	0.74*** (0.22)
5x+	0.27 (0.21)	0.63*** (0.16)	1.15*** (0.21)	0.90*** (0.26)
Period				
Constant	-4.41*** (0.13)	-2.47*** (0.057)	-3.91*** (0.11)	-5.71*** (0.21)
$t > 3$	2.91*** (0.25)	1.26*** (0.28)	2.56*** (0.27)	2.59*** (0.33)
$t > 5$	0.073 (0.21)	-0.35 (0.32)	-0.50 (0.30)	0.18 (0.28)
$t > 7$	0.059 (0.24)	-0.55 (0.37)	0.55* (0.29)	0.50* (0.28)
N	8897			8897

*The table reports coefficients from multinomial logit models for the action choice probabilities of the regulatory machine and the plant conditional on the state within the penalty stage. See Table S.1 for action definitions. Plant and regulatory actions are reported in administrative data by the regulator. The omitted action for the regulator is accept and for the plant is ignore, so the coefficients are to be interpreted as the effect of each component of the state on the party taking the specified column action relative to the omitted action. Pollution readings are taken during inspections throughout the treatment period. The omitted pollution reading is null, which occurs when the regulator inspects but does not take a pollution reading. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

(b) Table IV: Multinomial Logit Model of Action Choice Conditional on State

Read: Figure 4 lays out the dynamic penalty-stage game. The regulator observes pollution and can inspect, warn, punish, or accept. The plant can ignore or comply. Table IV estimates how the regulator and plants choose actions conditional on the state. High pollution readings make warnings and punishments more likely.

Economic message: The regulatory machine is not random noise. It uses information from past actions and pollution readings. However, the probability of costly punishment remains limited, which constrains the deterrence power of any one inspection.

8.5 Table V: Estimates of Plant Utility Parameters and Figure 5: Value of Environmental Regulation for Plants

2148

DUFLO, GREENSTONE, PANDE, AND RYAN

TABLE V
ESTIMATES OF PLANT UTILITY PARAMETERS (USD $\times 10^3$)^a

	Whether Bribes Given if No Compliance			
	No Bribes		On Inspection	
	(1)	(2)	(3)	(4)
<i>Parameters of Penalty Function</i>				
τ_0	53.54 (24.68)		28.12 (20.88)	36.71 (22.92)
τ_1		39.57 (28.17)		
τ_2		54.11 (27.43)		
τ_3		41.51 (19.15)		
<i>Parameters of Bribe Function</i>				
ν_0			9.67 (3.07)	
ν_1				10.93 (3.48)
ν_2				9.72 (3.99)
ν_3				5.83 (4.96)
<i>Standard Deviation of Action Shock</i>				
σ	5.02 (0.46)	5.88 (0.30)	5.11 (0.39)	4.83 (0.43)
Observations	1474	1474	1474	1474

^aThe table presents pseudo-maximum-likelihood estimates of the parameters of the plant profit function from the estimation of the plant's dynamic problem in the penalty stage. The four columns represent different specifications for the penalties and bribes the plant must pay. The parameters τ give the value of penalties applied by the regulator, by choosing the action punish, conditional on the pollution component of the state being between the standard and twice the standard (τ_0), between twice and five times the standard (τ_1), and above five times the standard (τ_2). The column 3 and 4 estimates also include estimates of bribes in addition to formal penalties. The parameters ν , for which estimates are reported in columns 3 and 4, give the value of bribes given by the plant in penalty specifications where the plant is assumed to give bribes, if it has not already complied in the stage, at the second inspection, and later inspections. The final parameter σ is the standard deviation of the plant's action-specific payoff shock. Observations are those at which the plant moves in rounds $t = 4$ and onward; $t = 2$ is omitted because a large number of actions in that round are imputed (see text). Inference is by the bootstrap over 100 samples with replacement, where samples are taken at the level of the plant chain (i.e., series of interactions) stratified on the maximum pollution reading observed in the chain. Standard errors equal to the standard deviation of bootstrap estimates are given in parentheses.

(a) Table V: Estimates of Plant Utility Parameters

THE VALUE OF REGULATORY DISCRETION

2149

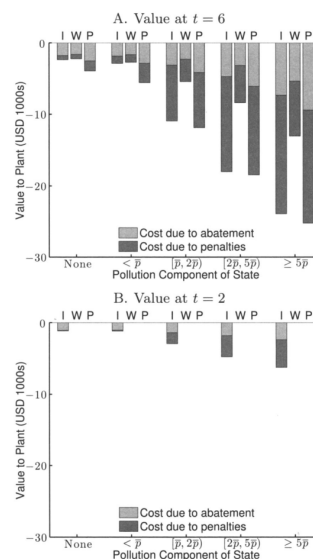


FIGURE 5.—Value of environmental regulation for plants. The figure shows the cost of regulation to plants in thousands of U.S. dollars as measured by the expected discounted value of different states in the penalty stage. Values are divided between expected discounted future abatement costs (light grey) and expected discounted future regulatory penalties (dark grey), both of which, as costs to the plant, have negative value. The figure shows three different dimensions of the state along which plant value varies. First, the panels show the time dimension evaluated when it is the plant's turn to move (A) at $t = 6$, and (B) at $t = 2$. Second, the five clusters of bars on the horizontal axis show different maximum lagged pollutant readings observed during the prior inspection. Third, within each group, the letters I, W, and P show how the value to the plant changed if the regulatory machine's lagged action was inspect, warn, or punish, respectively.

(b) Figure 5: Value of Environmental Regulation for Plants

Read: Table V estimates the monetary value of regulatory penalties and other costs. The estimates imply that punishments such as plant closure are expensive when applied. Figure 5 translates those costs into the value to the plant of being in different penalty-stage states. Values become more negative when pollution is high and the regulatory chain is late enough that punishment is more likely.

Economic message: The structural penalty estimates explain why targeted inspections can be powerful: extreme pollution readings create large expected costs. But the same estimates also explain why random inspections can be weak: for many moderate violations, expected penalties are too small to justify costly abatement.

8.6 Table VI: Estimates of Targeting Stage Parameters and Figure 6: Sensitivity of Targeting Parameters to Moments

THE VALUE OF REGULATORY DISCRETION

2151

TABLE VI
ESTIMATES OF TARGETING STAGE PARAMETERS*

	Constrained		Unconstrained	
	Initial Inspections (1)	Log Pollution (2)	Initial Inspections (3)	Log Pollution (4)
<i>Panel A. Targeting and Pollution Equations</i>				
Inspection treatment	0.095 (0.009)		0.162 (0.025)	
Run equipment (=1)		-1.902 (0.160)		-0.711 (0.308)
Inspection targeting shift parameter (λ_1)	-0.395 (0.003)		-0.220 (0.066)	
Inspection targeting level parameter (λ_2)	33.022 (1.876)		10.064 (3.137)	
Constant		0.212 (0.109)		-0.009 (0.102)
<i>Panel B. Distributions of Pollution and Maintenance Cost Shocks</i>				
Standard deviation of observed pollution shock (σ_1)	0.069 (0.003)		0.111 (0.022)	
Standard deviation of unobserved pollution shock (σ_2)	1.033 (0.047)		0.864 (0.042)	
Mean of log maintenance cost (μ_c)	2.388 (0.061)		1.833 (0.334)	
<i>Panel C. Test of Targeting Optimality Constraints</i>				
Distance metric test statistic χ^2_2		16.1039		
Test p -value		0.0003		

*The table reports parameters of the targeting stage of the model. The first two columns 1 and 2 report estimates from the constrained model where the regulator is constrained to target optimally based on observed pollution shocks, and columns 3 and 4 report estimates from the unconstrained model. Within each pair of columns, the first column (columns 1 and 3, respectively) reports the coefficients on the inspection equation and the second column (columns 2 and 4) reports coefficients on the pollution equation. Panel B reports estimates of the distributional parameters for pollution and cost shocks under each model. Panel C reports the results of a test of the constraints that require optimal targeting ((16)–(18) in Appendix C.3). The data set is a cross section of all sample plants. Inspections are calculated as the number of initial inspections per year for each plant over the course of the approximately 2-year experiment, and pollution is the maximum (over pollutants) logged standardized end-line survey pollution reading, where the standardization is in units of the regulatory standard for each pollutant (e.g., twice the pollutant-specific standard is a value of $\log(2)$). Both the inspection and the pollution models additionally include the observable characteristics of audit treatment assignment, audit $\times$ inspection assignment, an audit sample dummy, and region dummies (not reported). The parameters are estimated via generalized method of moments using a mix of analytic and simulated moments as described in Section 5. Standard errors are bootstrapped with $B = 100$ and $S = 5000$.

the measure introduced by Andrews, Gentzkow, and Shapiro (2017) to provide a sensitivity analysis. This *sensitivity* measures how a parameter estimate would change at the margin, given a change in one moment, holding fixed all other moments that underlie the estimation. Since our structural estimates rely on experimental variation, this method is appealing to build intuition for how the estimates would change if the experimental results had been different.

Figure 6 reports the sensitivity of select parameter estimates (across panels) to select moments (row headers within the panel). The length of each bar is the increase (solid

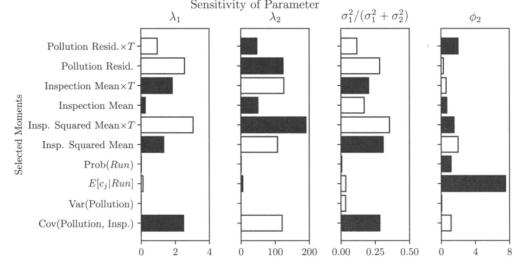


FIGURE 6.—Sensitivity of targeting parameters to moments. The figure shows the Andrews, Gentzkow, and Shapiro (2017) sensitivity measure of selected targeting model parameter estimates with respect to selected moments used to estimate the model. The panels, left to right, show the sensitivity of the parameters or functions of parameters λ_1 , λ_2 , and $\sigma_1^2/(\sigma_1^2 + \sigma_2^2)$, with respect to moments indicated by the row headers. The length of each bar is the local sensitivity of the parameter with respect to the row moment, measured in units of the parameter value per 1 standard deviation change in the moment. From the rows, we omit the products

(b) Figure 6: Sensitivity of Targeting Parameters to Moments

(a) Table VI: Estimates of Targeting Stage Parameters

Read: Table VI estimates the targeting and pollution stages. The constrained estimates imply aggressive targeting: the maximum inspection parameter is large, while the regulator observes only a small share of the total variation in pollution. Figure 6 shows which empirical moments most influence parameter estimates.

Economic message: The regulator appears to target strongly on weak information. This is a subtle but important point: discretion is useful, but the information on which discretion operates is noisy. The value of discretion would be larger with better monitoring data.

8.7 Figure 7: Observed and Optimal Targeting Rules and Figure 8: Value of Discretion: Abatement by Information Regime and Budget Constraint

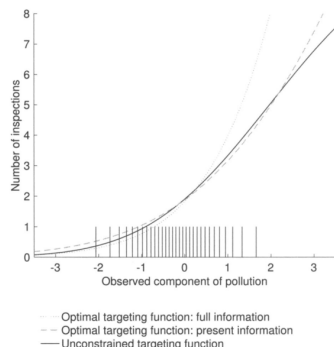


FIGURE 7.—Observed and optimal targeting rules. The figure shows several inspection targeting rules that assign plants an annual rate of inspection as a function of the component of pollution observed by the regulator. Two of the targeting rules are based on the estimates of Table VI: the dashed line gives the estimated optimal targeting function from the constrained estimates in the table, and the solid line gives the estimated targeting function from the unconstrained estimates. The dotted line gives the optimal targeting rule under an alternative regime where the regulator observes all the variance in pollution. The vertical spikes on the horizontal axis represent a normal distribution of pollution shocks, centered around the mean value of the observable characteristics $X_j^* \beta_1$, on which inspections are targeted.

(a) Figure 7: Observed and Optimal Targeting Rules

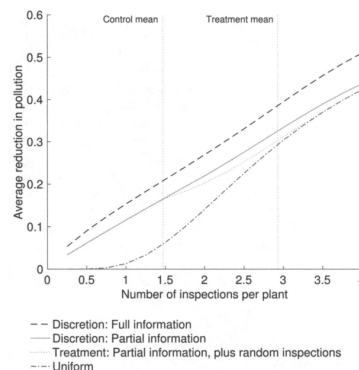


FIGURE 8.—Value of discretion: Abatement by information regime and budget constraint. The figure shows the amount of pollution abatement achieved, in units of regulatory standards of abatement per plant, under different counterfactual inspection regimes. Regimes vary in the information used by the regulator and the inspection budget available per plant. Each line shows average abatement per plant against the total inspection budget per plant per year (horizontal axis) under a different regime. The dashed and dotted line shows abatement under a minimum threshold rule that requires all plants to be inspected the same number of times. The solid line shows abatement under the optimal targeting rule where targeting is based on the observed component of pollution. The dashed line shows abatement under the same optimal targeting rule where the regulator is assumed to observe all variation in pollution. The dotted line is a hybrid regime, meant to reflect the experimental treatment, where inspections up to the control level of inspection (1.5 inspections per plant per year) are allocated with discretion, according to the optimal rule, and additional inspections beyond that level are allocated evenly across all plants.

(b) Figure 8: Value of Discretion: Abatement by Information Regime and Budget Constraint

Read: Figure 7 compares targeting functions under present information, full information, and unconstrained targeting. Figure 8 simulates abatement under alternative inspection regimes and budgets: uniform inspections, discretionary inspections with partial information, discretionary inspections with full information, and the experiment-like hybrid rule.

Economic message: Figure 8 is the paper’s policy punchline. At the control inspection budget, regulator-targeted inspections achieve roughly three times the abatement of uniform random inspections. Full information would further increase abatement, showing that the value of discretion depends on information quality.

9 Main findings by theme

9.1 Finding 1: the experiment sharply increased inspections

- Control plants received about 1.40 inspections per year during the treatment period.
- Treatment plants received about 3.11 inspections per year.
- The treatment increased initial inspections by about 1.50 per year.
- Perceived inspection rates also rose, so plants noticed the increased scrutiny.

9.2 Finding 2: more inspections produced more detected violations and citations

- Treatment plants were more likely to have a pollution reading collected.

- They were more likely to have a reading above the regulatory limit.
- They received more citations and more closure warnings.
- This shows that the inspection treatment was not merely procedural; it generated more regulatory information and formal violation records.

9.3 Finding 3: more inspections did not generate many more costly penalties

- Closure directions, bank guarantees, equipment mandates, and utility disconnections rose only slightly and insignificantly.
- Conditional on what an inspection found, the regulator followed up similarly in treatment and control groups.
- The weak penalty effect therefore comes from what random inspections found, not from treatment inspections being ignored.

9.4 Finding 4: average pollution effects were small

- The inspection treatment reduced standardized pollution by about 0.105 standard deviations, but this estimate is not statistically significant.
- Compliance increased by about 3.7 percentage points, significant at the 10 percent level.
- The increase in compliance occurs mainly near the threshold: plants just above the standard move just below it.

9.5 Finding 5: random inspections missed extreme violators

- The treatment found many additional plants above the standard.
- It did not find more plants with pollution five or ten times the standard.
- Extreme violators are exactly the cases most likely to trigger costly penalties.

9.6 Finding 6: the regulator targets aggressively but with weak information

- The model estimates imply very steep targeting: some plants are inspected frequently while many receive few inspections.
- The observable-to-regulator pollution signal explains only a small part of total pollution variation.
- This means discretion is valuable even with noisy information, but better signals would materially improve enforcement.

9.7 Finding 7: discretion has high value at low budgets

- At the observed control budget, targeted inspections generate about three times more abatement than uniform inspections.
- Adding random inspections to a discretionary baseline has a weak marginal effect because the added random inspections are spread thinly.

- If the added inspections had been targeted with the regulator’s discretion, the treatment effect on abatement would have been meaningfully larger.

10 Identification and interpretation

10.1 What the randomized experiment identifies

The experiment cleanly identifies the effect of moving treatment plants from the status quo to a regime with more inspections and random assignment of added inspections. The design is strong for measuring the effect of the bundled policy actually implemented.

10.2 What the experiment alone does not identify

The experiment alone cannot identify what would have happened if extra inspections had been allocated with discretion. It also cannot reveal whether the status quo discretionary allocation is optimal. These questions require the model.

10.3 Why the structural model is credible in this setting

- The model is disciplined by randomized variation in inspections.
- It uses detailed administrative data on the sequence of regulatory and plant actions.
- It matches key features of the inspection and pollution distributions.
- It is evaluated using sensitivity analysis that shows which moments matter for which parameters.

10.4 Why treatment effects are small despite more inspections

The key interpretation is selection. Under the status quo, the regulator used private information to inspect severe violators. The treatment added inspections randomly. Random inspection is useful for detecting general noncompliance, but it is relatively weak for finding high-penalty extreme cases. Since penalties are concentrated in extreme cases, random inspections generate limited additional abatement.

11 Economic intuition

11.1 Why targeted inspections can dominate random inspections

Suppose inspection budgets are scarce. If all plants are inspected equally, many inspections go to plants that are already clean or only mildly noncompliant. These plants may not face high enough expected penalties to justify costly abatement. If regulators can concentrate inspections on plants likely to be very dirty, expected penalties rise exactly where abatement yields the largest pollution reductions.

11.2 Why information quality matters

Discretion is only as good as the information behind it. If the regulator observes a strong signal of true pollution, targeted inspections can be very effective. If the signal is weak, some inspections will

be misallocated. The paper’s counterfactuals show that improving information can be as valuable as increasing inspection resources.

11.3 Why more inspections may have diminishing returns

At very low inspection budgets, discretion is especially valuable because the regulator must choose carefully. As budgets increase, even uniform inspections eventually reach many plants. The relative advantage of discretion declines as inspection resources become abundant.

12 Relation to broader literature

- **Environmental regulation.** The paper contributes causal evidence on the enforcement channel linking inspections, penalties, and emissions.
- **State capacity.** It shows that state capacity is not just the number of staff or inspections; it also includes information and allocation rules.
- **Bureaucratic discretion.** It provides empirical evidence that discretion can improve policy implementation when bureaucrats have useful local information.
- **Structural estimation with experiments.** It illustrates how a model can decompose a bundled treatment and simulate counterfactual policies not experimentally assigned.
- **Monitoring technology.** The findings motivate technologies such as continuous emissions monitoring systems because they improve the information input to discretionary enforcement.

13 Strengths and limitations

13.1 Strengths

- Field experiment implemented with a real regulator.
- Large sample of high-pollution-potential industrial plants.
- Rich administrative data on the enforcement process, not just outcomes.
- Independent end-line pollution measurements.
- Clear distinction between detection, citation, penalty, abatement, and emissions.
- Structural model tightly connected to the experiment and administrative process.

13.2 Limitations

- The treatment bundles more resources with reduced discretion, so the key decomposition depends on the structural model.
- Counterfactuals hold the penalty-stage policy fixed; in a long-run policy regime, regulators and plants might adapt.
- The model assumes a benevolent regulator in targeting, abstracting from corruption in assignment even though corruption may matter in some settings.

- End-line pollution readings are snapshots; plants may adjust behavior around visits.
- External validity depends on the regulator’s information, penalty credibility, plant technology, and institutional environment.

14 Short summary

- The paper studies environmental inspections in Gujarat, India.
- A randomized treatment increased inspection frequency and randomized the added inspections.
- Inspections, detected violations, and citations increased strongly.
- Costly penalties and average pollution reductions did not increase much.
- The treatment modestly improved compliance by moving some plants just below the legal standard.
- Random inspections did not find more extreme violators, suggesting that the regulator’s discretionary inspections were already targeted toward dirty plants.
- A structural model shows that discretion is valuable: regulator-targeted inspections produce much more abatement than random inspections at the same budget.
- The regulator’s information is weak, so better monitoring would substantially increase the effectiveness of discretion.

15 Conclusion

15.1 Core conclusion

Duflo, Greenstone, Pande, and Ryan show that increasing inspection resources is not automatically enough to improve environmental outcomes. The way inspections are targeted matters. Random additional inspections increased detection and citations, but generated little average emissions reduction because they did not find many extreme violators.

15.2 Incremental contribution revisited

The paper’s major contribution is to measure the value of regulatory discretion using a field experiment, rich administrative data, and a structural model. The experiment shows the effect of random additional inspections; the model uses that variation to infer how much abatement would be produced by alternative targeting regimes.

15.3 Policy implication

A regulator with discretion and partial information can outperform a uniform rule, especially when budgets are tight. However, the large gains from full-information counterfactuals show that discretion should be paired with better monitoring technology, not simply expanded without improving information.

15.4 Takeaway

The clean takeaway is: **resources, discretion, and information are complements**. More inspections help most when they are targeted well; targeting works best when the regulator observes good signals of pollution; and better information raises the return to both discretion and inspection budgets.